The Muffin Problem

Spring 2024 ARML Power Contest

The Solutions

1. Since there is only one student, no muffin needs to be cut, so the student receives only whole muffins.

2. One muffin needs to be divided fairly among s students, so each student gets a total of $1/s$ of a muffin. Since the one muffin can be divided into s evenly-sized pieces of $1/s$ each, no student needs to receive a piece smaller than that. So $f(1, s) = 1/s$.

3. By giving each student k uncut muffins every student receives their fair share of the muffins and no one needs to receive a piece that is smaller than a whole muffin. Thus $f(ks, s) = 1$.

4. To divide m muffins fairly among s students, each student receives m/s muffins. But since m is not a multiple of s, the number m/s is not an integer. Thus each student must receive some fraction of some muffin. In other words, at least one muffin needs to be cut into two of more pieces. If any piece of this muffin is larger than $1/2$ then the remaining pieces must total less than $1/2$ of a muffin. Thus, some student must receive a piece that is less than $1/2$ in size. Thus, if any muffin must be cut, the best that could be hoped for is if all muffins that must be cut will be divided into two equal pieces, so that no one receives a piece smaller than $1/2$. Thus $f(m, s) \leq 1/2$.

5. The numbers in a column of a muffin division table are the sizes of the pieces into which any particular muffin is cut. Reassembling these pieces gives one whole muffin, so the sum of the numbers in a column of the table is 1.

6. The numbers in a row of a muffin division table are the sizes of the pieces that each student receives from each muffin. The sum of these numbers is thus the total amount of muffin that the student receives. Since the muffins are divided fairly, this total must be a fair share of the muffins: m/s.

7. If a muffin is cut into three or more pieces, at least one of those pieces is $1/3$ of a muffin or less. Since the students have already found a division of the muffins where everyone receives no pieces that are smaller than $5/12$ and $5/12 > 1/3$, the "best" division of the muffins can't require a muffin to be cut into three or more pieces.

8. Since there are ten pieces of muffin to be apportioned amongst three students, at least one of the students must receive at least four pieces (this is one version of the pigeonhole principle). If a student receives four pieces of muffin totaling $5/3$ of a muffin, the smallest of those pieces cannot be larger than $\frac{5/3}{4} = 5/12$. Thus $5/12$ is the largest that the smallest piece anyone receives could be.

9. Here is one such muffin division table:

	m_1	m_2	m_3
s_1	$\frac{1}{4}$	$\frac{7}{20}$	
s_2	$\frac{1}{4}$	$\frac{7}{20}$	
s_3	$\frac{1}{4}$		$\frac{7}{20}$
s_4	$\frac{1}{4}$		$\frac{7}{20}$
s_5		$\frac{3}{10}$	$\frac{3}{10}$

10. If someone gets their fair 3/5 of a muffin in a single piece, then the muffin from which that piece was cut has 2/5 left over. If that piece is also given to a single person, then that person still needs 1/5 of a muffin to obtain their fair share. So all the other pieces of muffin they receive must be at most $1/5 < 1/4$ in size. Thus the smallest piece someone gets would be smaller than 1/4. On the other hand if the 2/5 leftover piece is cut into two or more pieces, then one of those pieces is at most 1/5 in size, and once again someone will receive a piece smaller than 1/4. So the best division of 3 muffins among 5 students cannot have someone receiving a single 3/5-sized piece.

11. Since everyone gets 3/5 of a muffin, if someone get three or more pieces, one of those pieces will be at most $\frac{3/5}{3} = 1/5 < 1/4$ in size. But no one gets any pieces smaller than 1/4, so no one gets three or more pieces.

12. Now we know that the muffins have been cut into exactly ten pieces. Using the pigeonhole principle, at least one of the muffins must be cut into at least four pieces. The smallest of those pieces must be no larger than 1/4. Thus in every division of three muffins among five students there is always at least one piece that is no larger than 1/4 in size. Since there is a division where the smallest piece is exaclty 1/4 as seen a few problems ago, the conclusion is that $f(3,5) = 1/4$.

13. Let $f(m,s) = a$ and $f(s,m) = b$. Since $f(m,s) = a$ there is a muffin division table for m muffins and s students where the smallest number in the table is a. Now multiply every number in the table by $\frac{s}{m}$. In this table, each row adds to $\frac{s}{m}\frac{m}{s} = 1$ and each column adds to $\frac{s}{m} \cdot 1 = \frac{s}{m}$. Transpose this table, making the rows into columns and the columns into rows. At the same time, exchange all the s and m labels. This relabeled table is now has every column adding to 1 and every row adding to s/m, so is a muffin division table for s muffins and m students. Thus, there is a way to divide s muffins among m students so that the smallest piece is $\frac{s}{m} \cdot a$. This shows that $b = f(s,m) \geq \frac{s}{m} \cdot a$. A similar argument shows that $a \geq \frac{m}{s} \cdot b$. Substituting,

$$b \geq \frac{s}{m} \cdot a \geq \frac{s}{m}\frac{m}{s} \cdot b = b.$$

Since the two ends of this inequality are equal, the inequalities must actually be equalities and the proof is complete.

14. If some muffin is cut into three or more pieces, than at least one of those pieces must be of size at most 1/3. Since $FC(m, s)$ is the larger of two values, one of which is 1/3, $FC(m, s) \geq 1/3$. So if the best way to cut the muffins requires some muffin be cut into three or more pieces, the upper bound is satisfied.

15. Since there are exactly $2m$ pieces to distribute to the s students, and m is not a multiple of s, there is at least one student who gets $\lceil 2m/s \rceil$ pieces or more, and there is at least one student who receives $\lfloor 2m/s \rfloor$ or fewer pieces.

    For a student who received at least $\lceil 2m/s \rceil$ pieces, then since they get a total of s/m muffins, their smallest piece can be no larger than $\dfrac{s}{m\lceil 2m/s \rceil}$. On the other hand for a student who received at most $\lfloor 2m/s \rfloor$ pieces, their *biggest* piece is *at least* $\dfrac{s}{m\lfloor 2m/s \rfloor}$ in size. The *other* piece from that muffin is of size *at most* $1 - \dfrac{s}{m\lfloor 2m/s \rfloor}$. Thus, *someone* receives a piece that is at most the smaller of $\dfrac{s}{m\lceil 2m/s \rceil}$ and $1 - \dfrac{s}{m\lfloor 2m/s \rfloor}$. In other words, if all muffins are cut into two pieces, some piece must be $\min\left\{\dfrac{s}{m\lceil 2m/s \rceil}, 1 - \dfrac{s}{m\lfloor 2m/s \rfloor}\right\}$ or less, and the proof that $FC(m, s)$ is an upper bound is complete in this case.

16. (a) When $m = 3k+1$ and $s = 3$ arithmetic shows that $2m/s = (6k+2)/3 = 2k+2/3$. So $\lceil 2m/s \rceil = 2k+1$ and $\lfloor 2m/s \rfloor = 2k$. Then

$$\begin{aligned} FC(3k+1, 3) &= \max\left\{\frac{1}{3}, \min\left\{\frac{3k+1}{6k+3}, 1 - \frac{3k+1}{6k}\right\}\right\} \\ &= \max\left\{\frac{1}{3}, \min\left\{\frac{3k+1}{6k+3}, \frac{3k-1}{6k}\right\}\right\} \end{aligned}$$

    Cross-multiplication shows $\frac{3k+1}{6k+3} > \frac{3k-1}{6k}$ and, since $k \geq 1$, $\frac{3k-1}{6k} \geq \frac{1}{3}$. So in this case, $FC(3k+1, 3) = \boxed{\dfrac{3k-1}{6k}}$.

    (b) When $m = 3k+2$ and $s = 3$ arithmetic shows that $2m/s = (6k+4)/3 = 2k+4/3$. This time, $\lceil 2m/s \rceil = 2k+2$ and $\lfloor 2m/s \rfloor = 2k+1$. Then

$$\begin{aligned} FC(3k+2, 3) &= \max\left\{\frac{1}{3}, \min\left\{\frac{3k+2}{6k+6}, 1 - \frac{3k+2}{6k+3}\right\}\right\} \\ &= \max\left\{\frac{1}{3}, \min\left\{\frac{3k+2}{6k+6}, \frac{3k+1}{6k+3}\right\}\right\} \end{aligned}$$

    Cross-multiplication shows $\frac{3k+2}{6k+6} < \frac{3k+1}{6k+3}$ (unless $k = 0$ when they are equal) and, since $k \geq 0$, $\frac{3k+2}{6k+6} \geq \frac{1}{3}$. So in this case, $FC(3k+2, 3) = \boxed{\dfrac{3k+2}{6k+6}}$.

    (c) Since $3k+1$ muffins are being split among three students, each student should get $k + 1/3$ muffins. Take $2k$ of the $3k+1$ muffins and divide them into two pieces of

sizes $\frac{3k-1}{6k} = FC(3k+1,3)$ and $\frac{3k+1}{6k}$. Take the remaining $k+1$ muffins and split them all in half (it is actually only necessary to split one of them in half when k is even and they can all be left intact if k is odd). Now give one student all $2k$ of the $\frac{3k+1}{6k}$-sized pieces, which comes out to $\frac{3k+1}{3} = k + 1/3$ muffins, which is their fair share. To each of the other students give k of the $\frac{3k-1}{6k}$ pieces (which totals $k/2-1/6$ muffins) and $k+1$ of the 1/2-size pieces. A quick bit of arithmetic shows that they now have $k/2 - 1/6 + (k+1)/2 = k - 1/6 + 1/2 = k + 1/3$ muffins.

(d) This time, each student should receive $k+2/3$ muffins. Split $2k+2$ of the muffins into pieces of sizes $\frac{3k+2}{6k+6}$ and $\frac{3k+4}{6k+6}$. The remaining k muffins should be split in half (this time, if k is even, they could all be left intact, while if k is odd only one of them needs to be split in half). Give one student all $2k+2$ of the $\frac{3k+2}{6k+6}$ pieces, which totals $\frac{3k+2}{3} = k+2/3$, their fair share. Given each of the other two students $k+1$ of the $\frac{3k+4}{6k+6}$-pieces and k of the 1/2-sized pieces. They have now received $\frac{3k+4}{6} + \frac{k}{2} = \frac{3k+4}{6} + \frac{3k}{6} = \frac{6k+4}{6} = k + 2/3$ muffins, and the division is complete.

17. A protocol to give each of five students 11/5 muffins is to split the muffins as follows: 8 muffins are divided into 13/30 and 17/30 pieces, 2 muffins into 14/30 and 16/30 pieces, and split the last muffin evenly in half. Then give two of the students four pieces of size 13/30 and a 14/30 piece, totaling $4 \cdot \frac{13}{30} + \frac{14}{30} = \frac{66}{30} = \frac{11}{5}$. Give two of the students thee pieces of size 17/30 and one of size $1/2 = 15/30$ totaling $3 \cdot \frac{17}{30} + \frac{15}{30} = \frac{66}{30} = \frac{11}{5}$. The final student gets the two remaining pieces of size 16/30 and the two remaining pieces of size 17/30, for a share of $2 \cdot \frac{16}{30} + 2 \cdot \frac{17}{30} = \frac{66}{30} = \frac{11}{5}$. Expressed as a muffin division table,

	m_1	m_2	m_3	m_4	m_5	m_6	m_7	m_8	m_9	m_{10}	m_{11}
s_1	$\frac{13}{30}$	$\frac{13}{30}$	$\frac{13}{30}$	$\frac{13}{30}$						$\frac{14}{30}$	
s_2					$\frac{13}{30}$	$\frac{13}{30}$	$\frac{13}{30}$	$\frac{13}{30}$	$\frac{14}{30}$		
s_3				$\frac{17}{30}$				$\frac{17}{30}$	$\frac{16}{30}$	$\frac{16}{30}$	
s_4	$\frac{17}{30}$	$\frac{17}{30}$	$\frac{17}{30}$								$\frac{15}{30}$
s_5					$\frac{17}{30}$	$\frac{17}{30}$	$\frac{17}{30}$				$\frac{15}{30}$

18. This challenge has two parts: find a procedure to divide 24 muffins among 25 students so that no one gets a pieces smaller than 8/25, and show that no procedure can produce a larger smallest piece.

The division procedure first. Take six muffins and divide them into 12/25- and 13/25-sized pieces. The next six will be divided into 11/25- and 14/25-sized pieces. The next six into 10/25- and 15/25-sized pieces. The last six muffins are divided into two pieces of size 8/25 and one of size 9/25.

Take the six 12/25-sized pieces, and give two pieces each to three students, so those three students now have 24/25 of a muffin. The six 13/25- and the six 11/25-sized pieces can be paired and given to the next six students. The 14/25- and 10/24-sized pieces go to the next six students, and six more students get one of the 15/25- and one

of the 9/25-sized pieces. That leaves 12 pieces of size 8/25; give three of those pieces to each of the four remaining students and the division has been accomplished.

Now, assume there exists a division of the muffins for which the smallest piece anyone gets is larger than 8/25. First note that no muffin can be left uncut, as that would give someone a whole muffin and $1 > 24/25$ which is the fair share. On the other hand, no muffin should be cut into four or more pieces, as then one of those would be at most $1/4 < 8/25$ in size. So all muffins should be cut into either two or three pieces.

Among the students, no one can get just one piece, because that piece would have to be of size 24/25 and the remaining pieces of that muffin are all less than 8/25 in size. On the other hand, if some student gets three or more pieces, then one of them is at most $\frac{24/25}{3} = 8/25$ and the assumption is that the smallest piece anyone receives is larger than that. Thus, each of the 25 students receives exactly two pieces of muffin.

That means there are 50 total pieces of muffin. Since each muffin is cut into either two or three pieces, to obtain 50 pieces requires that 22 muffins are cut into two pieces and 2 are cut into three pieces.

For the two muffins that are cut into three pieces, those pieces are all strictly between 8/25 and 9/25 in size. Larger than 8/25 by assumption, and smaller than 9/25 because if one piece is at least 9/25 then the other two pieces of that muffin total at most 16/25 so one of them is at most 8/25. Furthermore, *none* of the muffins that are cut into only two pieces can have one of those pieces in this range, because then the other piece of that muffin must be at least 16/25 in size, and for the student who has that piece, their *other* piece will be less than 8/25. So there are exactly six pieces of muffin whose size is strictly between 8/25 and 9/25.

Now consider the students who own those six pieces. Their *other* piece—recall that each student has exactly two pieces of muffin—will be strictly between 15/25 and 16/25. Furthermore, there can be no other pieces of muffin in this range because if there are, then for the student who owns such a piece, their other piece would be between 8/25 and 9/25 and those pieces are already all accounted for. So there are exactly six muffin pieces between 15/25 and 16/25.

Next, for each of those pieces, consider the other piece of the muffin from which it came (note that all the muffins that have been cut into three pieces have already been considered). Those pieces must all be strictly between 9/25 and 10/25 in size. Again, there cannot be any other pieces of muffins in that range, because the other piece of the muffin from which they come would be between 15/25 and 16/25 and again, those are already accounted for. So there are exactly six pieces of muffin with size between 9/25 and 10/25.

Continue in this fashion. Consider the students who own those pieces. Their other pieces are all strictly between 14/25 and 15/25 in size. Keep going back and forth between the two pieces owned by a student and the two pieces into which the muffins are split. The conclusion is that there are exactly six pieces of muffin of size in each of the intervals $(8/25, 9/25)$, $(9/25, 10/25)$, $(10/25, 11/25)$, $\ldots$, $(15/25, 16/25)$. There are eight such intervals, each containing six pieces of muffin, totalling 48 pieces of muffin.

Since there must be 50 pieces of muffin, a contradiction has been reached, showing that there is no division of 24 muffins among 25 students where the smallest piece anyone gets is more than 8/25.